

$M(1, \dots, n)$, color indices, take trace:

$$\text{tr}_R(T^1 \dots T^i M(\lambda_1, \dots, \lambda_n))$$

→ $M^{h_1, \dots, h_n}(p_1, \dots, p_n)$

Don't want to use Polarization vec's

$D=4$: $P_{\alpha\dot{\alpha}} := (\sigma^\mu p_\mu)_{\alpha\dot{\alpha}} = \begin{pmatrix} p_0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p_0 + p^3 \end{pmatrix}$

$$\det(p) = p_\mu p^\mu$$

Lorentz. $SL(2, \mathbb{C})$

* Massless:

$$\det(p) = 0$$

Thus p is a rank 1 so we can write

$$p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{2} \\ 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 0 \end{pmatrix}$$

* Real reps: $\tilde{\lambda}_\alpha = (\lambda_\alpha)^*$ (1, 3) sig

* We complexify: $\{P_{\alpha\dot{\alpha}}\}^{\mathbb{C}}$ and so

$$(i\mathbb{N}^4)^{\mathbb{C}}: G \cong SL(2, \mathbb{C}) \times \overline{SL(2, \mathbb{C})}$$

• For λ and $\tilde{\lambda}$ be real we are working in (2, 2) signature.

→ Null Space: $p_\mu p^\mu = 0$

Null Momenta:

(i) λ : two complex dim vec.

(ii) Phase invariance

$$\lambda_\alpha \rightarrow e^{i\theta} \lambda_\alpha$$

$$\tilde{\lambda}_{\dot{\alpha}} \rightarrow e^{-i\theta} \tilde{\lambda}_{\dot{\alpha}}$$

→ little group:

$$K_{\alpha\dot{\alpha}} = \lambda_{\alpha}^{(\kappa)} \tilde{\lambda}_{\dot{\alpha}}^{(\kappa)}$$

Define: $P = LK$

$$\lambda^{(P)} = L^{\beta}_{\alpha} \lambda_{\beta}^{(\kappa)}$$

$$\lambda_{\alpha}^{(P)} = \Lambda_{\alpha}^{\beta} \lambda_{\beta}^{(\kappa)} e^{i\theta(\Lambda, P)}$$

$$\langle \lambda_1, \lambda_2 \rangle = \epsilon^{\alpha\beta} \lambda_{\alpha} \lambda_{\beta} \equiv \langle 12 \rangle$$

$$[\tilde{\lambda}_1, \tilde{\lambda}_2] = \epsilon^{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_{\dot{\alpha}} \tilde{\lambda}_{\dot{\beta}} \equiv [12]$$

ex: $\langle 13 \rangle^4 / (\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle) \equiv \mathcal{H}$

In general $\lambda_{\alpha} \rightarrow t \lambda_{\alpha}$
 $\tilde{\lambda}_{\dot{\alpha}} \rightarrow t^{-1} \tilde{\lambda}_{\dot{\alpha}}$

so for 1: $\mathcal{H}' \sim \frac{t^4}{t^2} \mathcal{H} \sim t^2 \mathcal{H}$

since t signifies helicity $-1/2$

$$\mathcal{H}' \sim t^2 \mathcal{H}$$

Scales as a helicity -1 particle

* Amplitudes are not invariant.

The Brackets $\langle \cdot \rangle$, $[\cdot]$ are Lorentz inv.
 up to a phase generated by the
 little group.

$$\lambda_{\alpha}^{(P)} = \Lambda_{\alpha}^{\beta} \lambda_{\beta}^{(\kappa)} e^{i\theta(\Lambda, P)}$$

$$\Rightarrow M(t_{\alpha} \lambda_{\alpha}, t_{\alpha}^{-1} \tilde{\lambda}_{\dot{\alpha}}) = t_{\alpha}^{-2h_{\alpha}} M(\lambda_{\alpha}, \tilde{\lambda}_{\dot{\alpha}})$$

$$M(t_a \lambda_a, t_a \tilde{\lambda}_a) = t_a^{-2\eta_a} M(\lambda_a, \tilde{\lambda}_a)$$

$$(1, 1, 0, 0)$$

$$(-\frac{1}{2}, \frac{1}{2}, 1, i)$$

$$(-\frac{1}{2}, -\frac{1}{2}, -1, -i)$$

$$\lambda_{1\alpha} \tilde{\lambda}_{1\dot{\alpha}} + \lambda_{2\alpha} \tilde{\lambda}_{2\dot{\alpha}} + \lambda_{3\alpha} \tilde{\lambda}_{3\dot{\alpha}} = 0 = \sum p_\mu$$

Schouten Identity:

$$\langle ab \rangle \lambda_{c\alpha} + \langle ca \rangle \lambda_{b\alpha} + \langle bc \rangle \lambda_{a\alpha} = 0$$

Dot w/ $\lambda_{\dot{\alpha}}^{\dot{1}}$:

$$\langle 11 \rangle \tilde{\lambda}_{1\dot{\alpha}} + \langle 12 \rangle \tilde{\lambda}_{2\dot{\alpha}} + \langle 13 \rangle \tilde{\lambda}_{3\dot{\alpha}} = 0$$

$$\langle 11 \rangle = \langle \lambda, \lambda \rangle = \epsilon^{\alpha\beta} \lambda'_\beta \lambda'_\alpha = -\epsilon^{\alpha\beta} \lambda'_\alpha \lambda'_\beta = -\epsilon^{\beta\alpha} \lambda'_\beta \lambda'_\alpha$$

$$\boxed{\begin{array}{l} \text{So } \langle nn \rangle = 0 \\ \forall n \in \mathbb{Z} \end{array}}$$

$$\lambda'^\alpha \lambda'_\alpha = -\lambda'^\alpha \lambda'_\alpha$$

$$\lambda'^2 = 0$$

$$\langle 12 \rangle \tilde{\lambda}_{2\dot{\alpha}} + \langle 13 \rangle \tilde{\lambda}_{3\dot{\alpha}} = 0$$

Suppose $\langle 12 \rangle$ and $\langle 13 \rangle$ as well as $\tilde{\lambda}_{2\dot{\alpha}} \neq 0$. Then

$$\tilde{\lambda}_{2\dot{\alpha}} \propto \tilde{\lambda}_{3\dot{\alpha}} \Rightarrow \tilde{\lambda}_{2\dot{\alpha}} = k \tilde{\lambda}_{3\dot{\alpha}}$$

$$\langle 12 \rangle k + \langle 13 \rangle = 0$$

$$k = -\langle 13 \rangle / \langle 12 \rangle$$

• Conserve Momenta via:

(i) All $\hat{\lambda}$ are proportional

i.e. $[\hat{\lambda}_i, \hat{\lambda}_j] = 0$

$$\langle \lambda_i \lambda_j \rangle \neq 0$$

(ii) otherway around

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